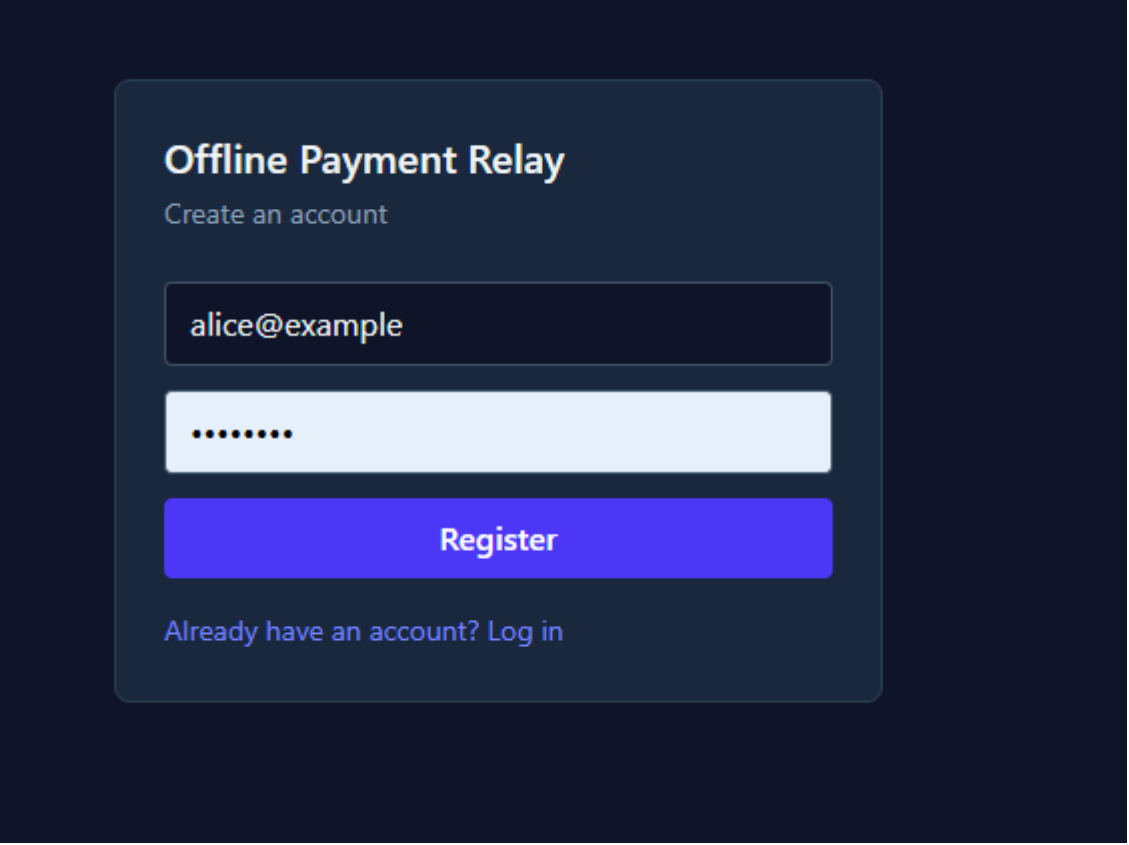
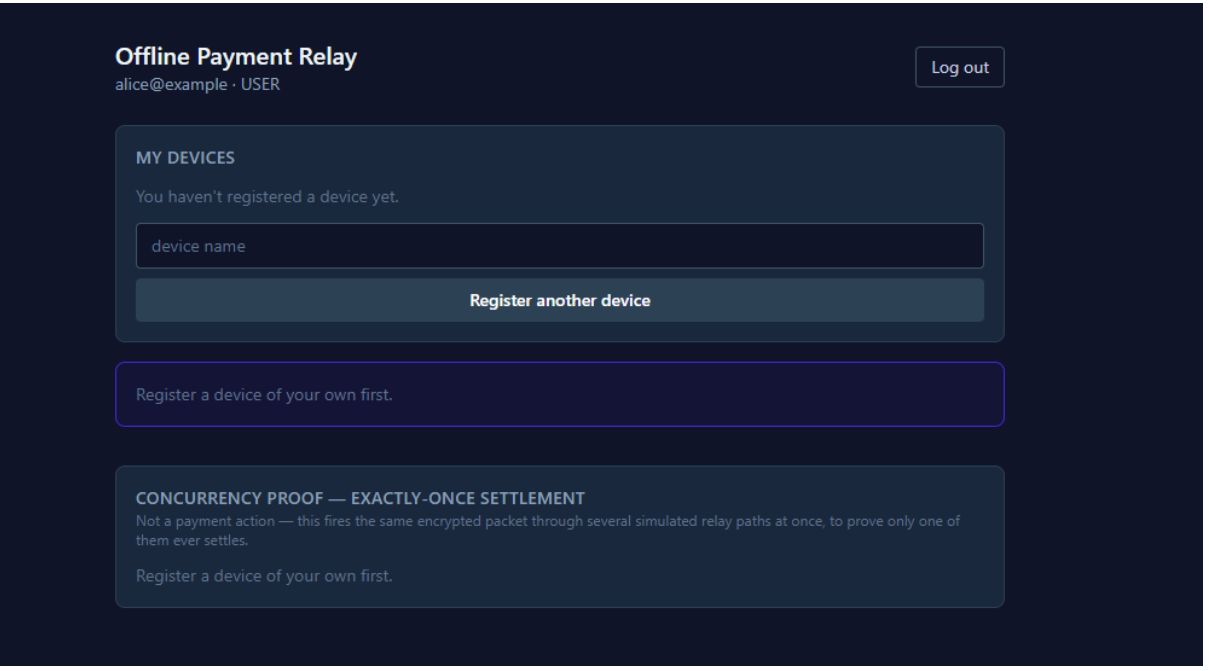


1) Register User 1 (Alice)



The image shows a registration screen for 'Offline Payment Relay'. The title 'Offline Payment Relay' is in a large, bold, light blue font. Below it, the subtitle 'Create an account' is in a smaller, lighter blue font. There are two input fields: the first contains the email 'alice@example' and the second contains a masked password '.....'. A prominent red button labeled 'Register' is centered below the fields. At the bottom, there is a link that says 'Already have an account? Log in'.

2) This is the screen that appears when users login. Enter a device name and register to get a unique device id that will be used for offline payments.



The image shows a user dashboard for 'Offline Payment Relay'. The header includes the title 'Offline Payment Relay' and the user's email 'alice@example' followed by 'USER'. A 'Log out' button is in the top right. The main section is titled 'MY DEVICES' and contains the message 'You haven't registered a device yet.' Below this is a 'device name' input field and a red button labeled 'Register another device'. A large red banner below the devices section says 'Register a device of your own first.' At the bottom, there is a section titled 'CONCURRENCY PROOF — EXACTLY-ONCE SETTLEMENT' with explanatory text and another 'Register a device of your own first.' message.

3) Copy the device id for future transactions. User can register multiple devices. Their default balance is \$0 and only admins can seed test balance into user accounts.

Offline Payment Relay

alice@example · USER

Log out

MY DEVICES

Alice's Iphone

487c1b9e-7ebd-4706-94ef-2c2d57c94b3a

Copy

Device name

Register another device

SEND PAYMENT

recipient device ID

paste the recipient's device ID

Look up

ALICE'S IPHONE WALLET

\$0

100.00

Seed test balance (admin)

CONCURRENCY PROOF — EXACTLY-ONCE SETTLEMENT

Not a payment action — this fires the same encrypted packet through several simulated relay paths at once, to prove only one of them ever settles.

recipient device ID

paste the recipient's device ID

Look up

4) Register User 2(Bob)

Offline Payment Relay

Create an account

bob@example

.....

Register

Already have an account? Log in

5) Register User 2's device in the same manner like User 1.

The screenshot displays the 'Offline Payment Relay' interface for a user named 'bob@example - USER'. It features a 'Log out' button in the top right corner. The main content is organized into four sections: 1. 'MY DEVICES' which lists 'Bob's Iphone' with a device ID 'efb19cd7-2b4b-444b-ba49-28352e844a12' and a 'Copy' button, followed by a 'device name' input field and a 'Register another device' button. 2. 'SEND PAYMENT' which includes a 'recipient device ID' input field with placeholder text 'paste the recipient's device ID' and a 'Look up' button. 3. 'BOB'S IPHONE WALLET' showing a balance of '\$0', an input field for '100.00', and a 'Seed test balance (admin)' button. 4. 'CONCURRENCY PROOF — EXACTLY-ONCE SETTLEMENT' with a descriptive note and another 'recipient device ID' input field with a 'Look up' button.

6) Login as admin

The screenshot shows the login page of the 'Offline Payment Relay'. It prompts the user to 'Log in to continue'. The login form contains an email input field with 'admin@demo.local', a password input field represented by dots, and a prominent blue 'Log in' button. Below the login fields is a link that says 'Need an account? Register'.

7) Use the admin login to seed an amount to User 1

Before:

Offline Payment Relay

admin@demo.local · ADMIN

Log out

MY DEVICES

admin

72cf1782-2ef7-4bca-bf71-6bcd623b2707

Copy

device name

Register another device

SEND PAYMENT

recipient device ID

487c1b9e-7ebd-4706-94ef-2c2d57c94b3a

Look up

Sending to: Alice's Iphone

amount

25.00

Send

ADMIN WALLET

\$100

100.00

Seed test balance (admin)

CONCURRENCY PROOF — EXACTLY-ONCE SETTLEMENT

Not a payment action — this fires the same encrypted packet through several simulated relay paths at once, to prove only one of them ever settles.

recipient device ID

paste the recipient's device ID

Look up

After: Balance after deduction

Offline Payment Relay

admin@demo.local · ADMIN

Log out

MY DEVICES

admin

72cf1782-2ef7-4bca-bf71-6bcd623b2707

Copy

device name

Register another device

SEND PAYMENT

recipient device ID

487c1b9e-7ebd-4706-94ef-2c2d57c94b3a

Look up

Sending to: Alice's Iphone

amount

25.00

Send

Sent \$25 — packet settled.

ADMIN WALLET

\$75

100.00

Seed test balance (admin)

CONCURRENCY PROOF — EXACTLY-ONCE SETTLEMENT

Not a payment action — this fires the same encrypted packet through several simulated relay paths at once, to prove only one of them ever settles.

recipient device ID

paste the recipient's device ID

Look up

8) Confirm by logging back in User 1's account. Ledger entry displays past transactions.

Offline Payment Relay

alice@example - USER

Log out

MY DEVICES

Alice's Iphone

487c1b9e-7ebd-4706-94ef-2c2d57c94b3a

Copy

device name

Register another device

SEND PAYMENT

recipient device ID

paste the recipient's device ID

Look up

ALICE'S IPHONE WALLET

\$25

100.00

Seed test balance (admin)

LEDGER

25 → balance 25 (3:37:02 PM)

CONCURRENCY PROOF — EXACTLY-ONCE SETTLEMENT

Not a payment action — this fires the same encrypted packet through several simulated relay paths at once, to prove only one of them ever settles.

recipient device ID

paste the recipient's device ID

Look up

9) Trying to pay over the current balance throws error.

Offline Payment Relay

alice@example - USER

Log out

MY DEVICES

Alice's Iphone

487c1b9e-7ebd-4706-94ef-2c2d57c94b3a

Copy

device name

Register another device

SEND PAYMENT

recipient device ID

efb19cd7-2b4b-444b-ba49-28352e844a12

Look up

Sending to: Bob's Iphone

amount

35.00

Send

sender wallet has insufficient balance

ALICE'S IPHONE WALLET

\$25

100.00

Seed test balance (admin)

LEDGER

25 → balance 25 (3:37:02 PM)

10) Using User 2's device ID send a valid amount. New balance shown and packet settled notification shown.

Offline Payment Relay

alice@example · USER

Log out

MY DEVICES

Alice's Iphone

487c1b9e-7ebd-4706-94ef-2c2d57c94b3a

Copy

device name

Register another device

SEND PAYMENT

recipient device ID

efb19cd7-2b4b-444b-ba49-28352e844a12

Look up

Sending to: Bob's Iphone

amount

15

Send

Sent \$15 — packet settled.

ALICE'S IPHONE WALLET

\$10

100.00

Seed test balance (admin)

LEDGER

25 → balance 25 (3:37:02 PM)

Reflects on device 2 as well:

Offline Payment Relay

bob@example · USER

Log out

MY DEVICES

Bob's Iphone

efb19cd7-2b4b-444b-ba49-28352e844a12

Copy

device name

Register another device

SEND PAYMENT

recipient device ID

paste the recipient's device ID

Look up

BOB'S IPHONE WALLET

\$15

100.00

Seed test balance (admin)

LEDGER

15 → balance 15 (3:39:32 PM)

Proof of concurrency test:

The console lets you test concurrency by letting you fire N different packets to the same account.

- 1) Select an amount, device ID and number of requests.

BOB'S IPHONE WALLET

\$15

Seed test balance (admin)

LEDGER

15 → balance 15 (3:39:32 PM)

CONCURRENCY PROOF — EXACTLY-ONCE SETTLEMENT

Not a payment action — this fires the same encrypted packet through several simulated relay paths at once, to prove only one of them ever settles.

recipient device ID

Look up

Target: Alice's Iphone

amount

concurrent paths

Simulate duplicate delivery

- 2) The console shows which path is chosen and the payment is settled.

CONCURRENCY PROOF — EXACTLY-ONCE SETTLEMENT

Not a payment action — this fires the same encrypted packet through several simulated relay paths at once, to prove only one of them ever settles.

recipient device ID

Look up

Target: Alice's Iphone

amount

concurrent paths

Simulate duplicate delivery

5 paths delivered, 1 settled, 4 blocked

sim-path-1
BLOCKED

sim-path-2
BLOCKED

sim-path-3
BLOCKED

sim-path-4
SETTLED

sim-path-5
BLOCKED